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RESEARCH ARTICLE

A Conversational AI Bot for Efficient Learning: A Prototypical Design

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ABSTRACT In an ideal world, how individuals learn should align with the methods they are taught. However, this is not always the case, and it is necessary to reconsider the approach to teaching in academic institutions, such as universities, to help students learn at their own pace. Innovative technology plays a vital role in achieving this objective. This study proposes using Artificial Intelligence (AI) to teach in academic institutions. Specifically, an AI-powered conversational bot called Conversational AI Bot for Efficient Learning (COFFEE) is designed, implemented, and will be used as an adaptive learning technology. COFFEE will analyze students' learning habits and adjust the teaching style based on their strengths and weaknesses. Teaching and learning are closely linked to the disciplinary areas of education, behavior, and technology. Combining these areas into a conversational bot equipped with AI algorithms and access to big data makes it possible to disrupt traditional teaching methods. This will benefit students who require special attention, such as those with severe learning disabilities who cannot receive adequate support from instructors due to time constraints or lack of experience. COFFEE is a valuable tool that can be used in academic institutions at all levels, including primary and secondary education. It will be based on the latest advancements in AI and data science and is designed to be feasible, reliable, and customizable.

INDEX TERMS Conversational systems, artificial intelligence, adaptive learning, severely challenged learners.

I. INTRODUCTION

The 2030 Sustainable Development Goals (SDG) postulate an international framework for dignity, peace, and prosperity for people and the planet, both now and in the future [1]. Since its inception in 2015, countries have translated this shared vision into national development plans and strategies. The SDG includes 17 Sustainable Development Goals, with Goal 4 being to ''Ensure inclusive and equitable quality education and promote lifelong learning opportunities for all.'' This goal also emphasizes free primary and secondary education, access to updated educational facilities, and instruction from competent teachers. Certain countries have encountered more difficulty meeting these goals in recent years than others. A United Nations (UN) report indicated that more than half of children and adolescents fail to achieve minimum proficiency in reading and mathematics. The report also highlights the need for more trained teachers to ensure quality education. The world requires more qualified teachers to meet the SDG above. An estimated 20 million primary and secondary school teachers must be added to the workforce, along with replacements for approximately 48 million expected to leave over the next 13 years due to retirement, the end of temporary contracts, or the desire to pursue a different profession with better pay or improved working conditions.

In connection with the education domain, the study in [2] reported that blended or hybrid learning, which combines face-to-face and online learning, is the most promising approach for higher education, making teaching and resource allocation more efficient. In addition, Gleason

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and Greenhow [2] reported that Social Robotic Telepresence Systems (SRTSs) offer advantages over traditionally used videoconferencing, including fostering students' embodiment in the classroom, enhancing their feelings of belonging and trust, and improving their ability to contribute ideas authentically. These advances in Artificial Intelligence (AI) enable new models of hybrid education to support social interactions, allowing learners to study in different locations and at different times through collaborative distance learning. This explains why the number of universities using such types of blended education is increasing [3]. Nouman et al. [4] reported that the number of affected learners constituted 68% of the total enrolled learners worldwide due to the closure of educational institutes during COVID-19, contributing to the growing demand for blended or e-learning delivery models.

Al Faisal et al. [5] reported statistics about childhood disability among the student population. According to the study, the considered disabilities include hyperactivity, mental disability, visual disability, autism, Attention Deficit Hyperactivity Disorder (ADHD), Down syndrome, dwarfism, speech delay, and developmental delay. Therefore, innovative technologies must be employed to develop tools, such as bots, for adaptive learning. The developed tools should help students with disabilities engage in the learning process as regular students.

Although conversational bots have been deployed in various domains, such as commerce and transportation [6], they have received limited attention in education. Tarek et al. [7] proposed a chatbot based on the chatbot development framework Xatkit. The proposed bot aims to assist teachers and students; however, it was presented conceptually without real-time evaluations. Therefore, its performance information needs improvement. Corral [8] examined the use of artificially intelligent chatbots for educating healthcare professionals.

Therefore, investigating how an AI-enabled tool such as COFFEE can help address the gaps between students and teachers is interesting. The AI-enabled bot, COFFEE, will not require days off and will never be late for work. Administrators can upload any changes to curricula across an entire fleet of teaching bots, and the systems will remain error-free. In addition, when programmed correctly, they will not exhibit biases toward students based on gender, race, or socioeconomic status. Anthony Seldon, a British education expert, made a strong statement at the British Science Festival in September 2017, asserting that robots will replace teachers by 2027 [9].

COFFEE will be unique and different from any other bot on the market, as it will be customizable to several education curricula, designed for both students and teachers, and can be further developed to support courses delivered in various contexts and at different levels.

A. RESEARCH CONTRIBUTIONS

New technologies, such as AI's conversational bots, can improve education. When fully exploited, these technologies will change how teaching is conducted, how learning occurs, and how the educational process is administered. The bot can help address the challenge of connecting mass-production educational systems to personalized learning environments, providing a high-quality learning experience. This can be achieved through adaptive learning scenarios.

This research seeks to understand how bots can support the teaching of students in academic institutions, such as universities, allowing them to learn at their own pace. The goal is to design and implement an AI conversational bot for efficient learning. The bot represents a stepping stone toward utilizing AI as an adaptive learning technology in teaching at academic institutions. The contributions of this study are outlined as follows:

- Conceptual design of an AI-enabled bot (COFFEE) based on cutting-edge techniques and tools. In addition, the conceptual design addresses the challenge of integrating different technologies into one AI-based bot.
- Implementation of the COFFEE bot using various technologies available in the market, with particular attention given to adaptive learning and suitability for students with special needs.
- Development of a complete evaluation environment to test and assess the functionalities of the AI-enabled bot.
- Identification of generalization and design patterns as part of the design science methodology paves the way for further research and development by other scholars.

B. PAPER STRUCTURE

The rest of this paper is structured into several parts, beginning with a brief analysis of the current state of the art in Section II. Section III introduces Design Science Research (DSR) as the methodology used in this research. The actual design and implementation processes are explained in Section IV. Section V describes the design patterns of the chatbot as part of the DSR methodology. Section VI presents the obtained results, whereas Section VII briefly discusses and highlights the research limitations. Finally, conclusions are drawn in Section VIII.

II. RELATED WORK

In 2021, the work in [10] offered a comprehensive literature review on chatbot applications in education, where different mechanisms and techniques were studied and presented. The authors argue that chatbots support teaching and learning activities, meaning chatbots can bridge gaps for teachers and students.

Ericsson et al. [11] explored Swedish school students' experiences interacting with an Embodied Conversational Agent (ECA). They investigated the cognitive, emotional, and social dimensions of the students' experiences and identified factors that can influence these experiences. They employed a multi-method approach, using data from a digital logbook and a post-task questionnaire. The logbook was based on the development of Experience Sampling, while the

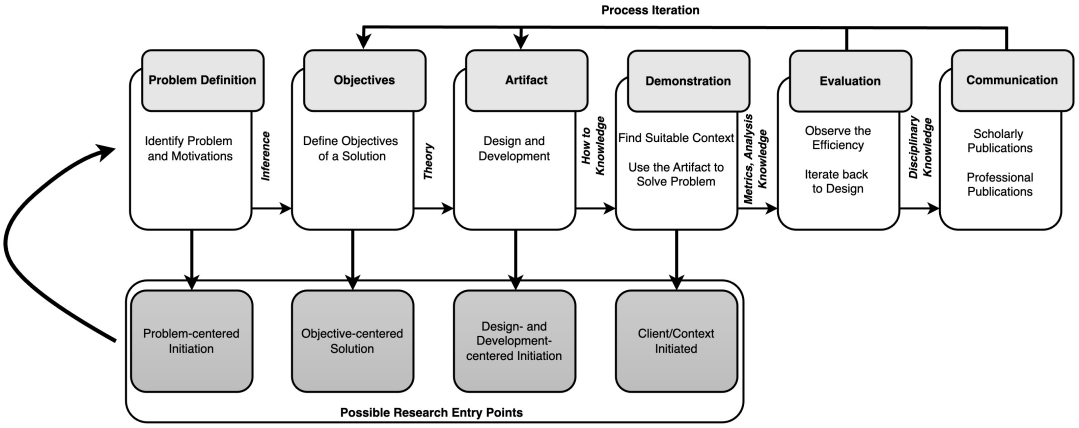


FIGURE 1. Design Science Research (DSR) methodology. The figure is excerpted from [12].

questionnaire was constructed in Swedish and adapted to the age group. The data were analyzed using descriptive statistics and thematic analysis. The results indicated that the students generally found it easy to communicate with and understand the ECA, which was effective with their language learning goals. The study emphasized the potential of ECAs as a tool for enhancing language learning and suggested that further research is needed to explore the impact of this technology in different contexts.

Kanodia et al. [13] focused on designing and implementing a robust Question Answering Model (QAM) using the Dialogflow platform and the BERT large language model, communicating via a Python Flask API-based webhook. They built the QAM to answer questions on various topics. The conversational chatbot employs BERT to facilitate accurate responses to user queries from extensive datasets. Users can input a document or a passage, and the QAM evaluates it to extract an answer, enhancing interactive dialogue. A crucial part of this implementation is categorizing user questions for higher response accuracy, addressing BERT's constraints on text string length. The fine-tuned model, trained and tested on the CoQA dataset, displays 98% and 96% accuracy rates, respectively. The user-friendly design of the conversational chatbot is also a crucial aspect of the project, enhancing its overall utility. Dialogflow communicates with the BERT model through a fulfillment service, webhook, and Flask API to deliver answers, making this chatbot an efficient and interactive solution for question-answering tasks.

Xie et al. [14] introduced a conversational agent, the Motivational Online Conversational Agent (MOCA), designed to increase student engagement and motivation in Computer-Supported Collaborative Learning (CSCL) environments. MOCA applies principles of motivational interviewing and employs an intelligent dialog engine to target students with low participation rates in CSCLs. In addition, MOCA implements multi-turn conversation interventions to encourage positive changes in student engagement. It was tested with 40 volunteers to evaluate MOCA's effectiveness, substantially increasing student participation scores. Although the study lacks precise implementation details, it outlines MOCA's architecture, which includes modules for information input, natural language comprehension, dialog management, answer generation, and feedback. The MOCA Dialogflow begins by greeting students and gathering basic information, followed by guiding student attitudes toward their potential intent for change, aiming to enhance motivation.

Several attempts to design chatbots for educational purposes have been identified, such as in [15], [16], [17], [18], and [19]. These studies confirm the feasibility of the conversational bot concept. In contrast, they need improvement, particularly in addressing limitations such as limited capabilities, support for only a particular language, and reliance on a question-answer learning model.

III. RESEARCH METHODOLOGY

Information Systems (IS) are an area of research that is concerned with the design of artifacts [20], and accordingly, they have become one of the field's main research points [21]. Reference [22] defined Design Science Research (DSR) as ''research that invents a new purposeful artifact to address a generalized type of problem and evaluates its utility for solving problems of that type.' A focal point of DSR is to develop design knowledge, which can be represented in different forms and at various levels of abstraction [23], such as constructs, models, methods, design principles, or design theories [24], [25], [26]. This study employed DSR as the methodology to answer the research question by designing and evaluating process models to enable organizations to use external data for data-driven decision-making. Hence, the [12] DSR methodology is adopted, which consists of six phases: problem identification, solution objectives, design and development, demonstration, evaluation, and communication, as shown in Figure 1. The DSR methodology

TABLE 1. Concepts and descriptions used in the chatbot design process.

Concepts	Description
Stakeholders	A stakeholder is a person or an organization that is influenced by the work of the DSR research outcomes [27]. This research identifies instructors, students, designers, and operators as stakeholders.
Needs	The need is sometimes seen as equivalent to the problem. Besides, the needs can also be broader than the problem as it might entail opportunities besides the problem to be solved. Therefore, the research project must solve the problem of the inability of learning management systems to provide an efficient learning environment to all classes of learners, let alone those severely challenged.
Goals	The goals represent the desired results or a desired state of affairs. This research project seeks to increase the overall quality of learning and the learning environment at academic institutions via AI technologies, especially conversational systems.
Requirements	Based on IEEE standards, a software requirement can be defined as: 1) A condition or capability needed by a user to solve a problem or achieve an objective 2) A condition or capability that must be met or possessed by a system, system component, product, or service to satisfy an agreement, standard, specification, or other formally imposed documents 3) A documented representation of a condition or capability as in (1) or (2) 4) A condition or capability that must be met or possessed by a system, product, service, result, or component to satisfy a contract, standard, specification, or other formally imposed document This research identifies the requirements via relevant literature and informal interviews with relevant stakeholders as follows: 1) Design a working prototypical implementation of an AI-based conversational system for efficient learning 2) Implement the design via IT artifacts.

described in this study is instantiated in the following subsections using the above steps [12].

A. PROBLEM IDENTIFICATION AND CONCEPTUALIZATION

This research examines the plausibility of the assumption that it is possible to design a conversational system capable of acting as an extension to the learning management system, but with extended features, such as mobility, adaptivity, and the ability to provide material in an environment that can adapt to changes required to meet learners' varying levels of severity. Academic institutions face many challenges in this area. There must be awareness and a willingness to provide an efficient learning environment that accommodates learners' needs. This task carries an increased risk, as investigations cannot yield the desired results. Identifying and addressing severity levels through adequate and appropriate technologies is part of the challenge. However, the potential utility is significant, as this technology can provide a way to include all types of learners with various abilities and disabilities. Design science research produces innovative solutions to real-world problems. DSR introduces systems' artifacts and design knowledge that describe the relationships between problem and solution spaces. An appropriate understanding and description of the underlying problem space is a foundational element of DSR research. This study conceptualizes the problem space using four key concepts: stakeholders, needs, goals, and requirements [27]. Table 1 describes the four fundamental concepts used in this study.

B. DEFINE OBJECTIVES OF THE SOLUTION

This research project seeks to utilize conversational AI bots to support learners at academic institutions. The goal is to design and implement an AI conversational bot for efficient learning. This study goes beyond the current state of the art by providing a learning environment suited to the needs of severely challenged learners.

C. DESIGN AND DEVELOPMENT

The proposed design is presented in the conceptual design shown in Figure 2. At its core is a conversational AI bot designed to interact with learners. Google Dialogflow (GDF) will be the platform used in this study. The bot retrieves course material from a Learning Management System (LMS). Canvas is the LMS used in this work. Some learner activities are logged in a database; therefore, the conceptual design includes a database component for tracking and reporting. Since some learners can have disabilities, the bot should be capable of interacting with those learners through an accessibility framework. Infusion was selected as the framework in this study. Learning is intended to be adaptive, so this work is implemented adaptivity at the assessment level. Mastery Path is the tool employed to implement adaptivity in this research.

D. THE INFUSION FRAMEWORK

The Infusion framework was used for the bot's interface. This choice is supported by its ability to provide an

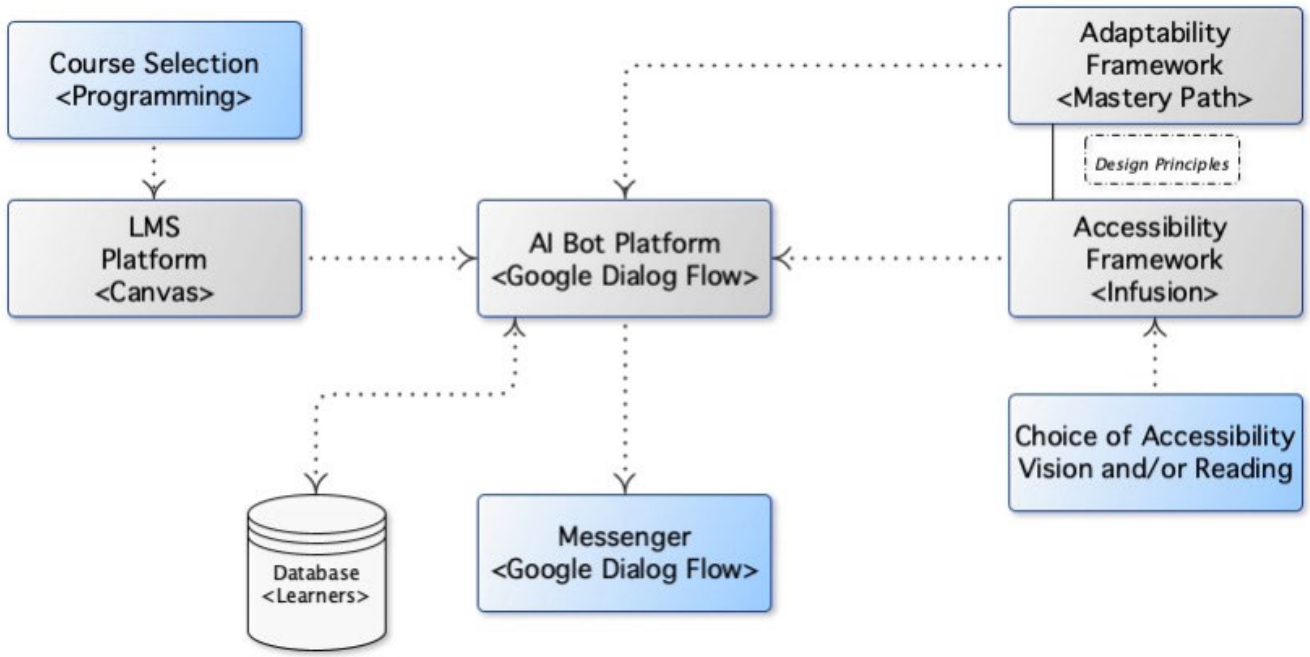


FIGURE 2. Conceptual design of the proposed chatbot.

accessible interface for individuals with reading disabilities. The Infusion framework offers reusable programming APIs and UI building blocks to assist in designing, developing, and integrating preference editors in web-based applications, content management systems, and content delivery environments. It enables the creation of user interface templates in HTML and allows the pages to be rendered entirely on the client side.

E. MASTERY PATHS ADAPTIVE ASSESSMENT

Adaptive learning facilitates differentiated learning experiences for students. Mastery Paths is used for that purpose in this research. Mastery Paths allows instructors to customize learning pathways for students based on their performance.

Mastery Paths enables instructors to configure differentiated learning content automatically delivered to students based on their course assignment scores. Any graded assignment, discussion, or quiz can be the source content for Mastery Paths. After the source assignment is graded, Mastery Paths sends content to students based on the rules governing their scoring ranges. A student's score determines their assigned learning path. For instance, a student who scores 80% or more on an assignment is assigned advanced work, while a student scoring below 65% is directed to assignments designed for content clarification and improvement. In this way, Mastery Paths is utilized to achieve adaptive learning.

F. DEMONSTRATE

The conceptual design is demonstrated through use case scenarios. These case scenarios involve instructors, students, designers, and operators as users. They work with different systems to complete various tasks. For example, both instructors and students use the LMS. Likewise, the bot platform is used by different user roles on different occasions. Supporting technologies, such as database management systems and prediction libraries, are used in the backend to support front-end functionalities and features. For further details, refer to the use cases depicted in Figure 3.

G. EVALUATE

A system with GDF at its core and several other technologies was implemented to evaluate the proposed IT artifact designed in this study. These include Canvas as the LMS, PostgreSQL as the database engine, Python libraries for machine learning prediction models, Tableau for result visualization, ChatGPT to answer questions beyond the course material, Infusion as the accessibility framework, and Mastery Path as the adaptive assessment tool. The evaluation environment is illustrated in Figure 4.

IV. IMPLEMENTATION

This section describes the prototype instantiation. It outlines the assumptions made, the procedure followed, and the technology used, concluding with a reflection on the results attained.

The assumptions are the following: a learning management system exists; course material exists; an assessment is utilized to adapt study material to the level of learners (a quiz is a mechanism for evaluation); students have varying levels of severity regarding vision, which influences their ability to

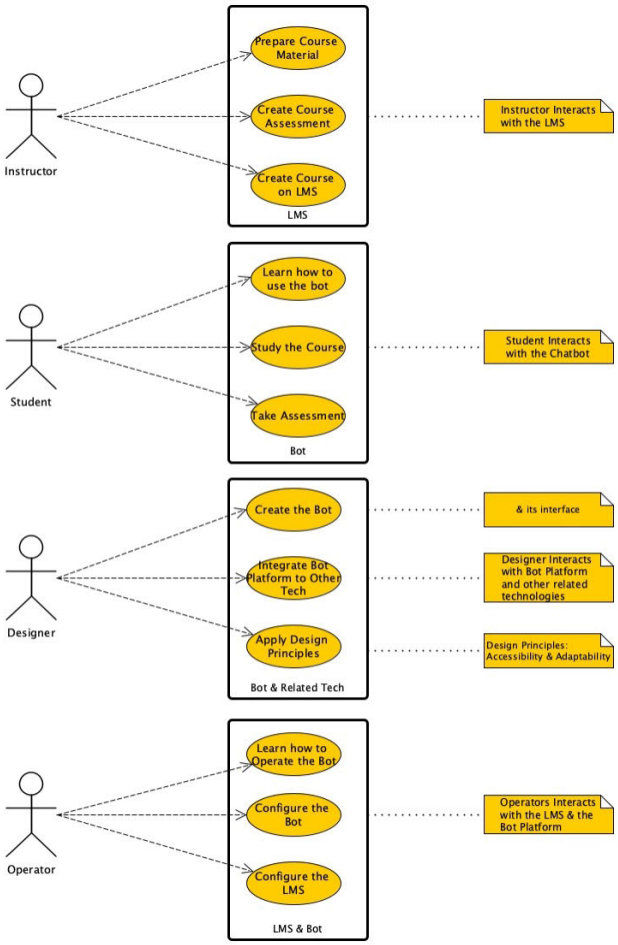


FIGURE 3. Use case scenarios.

read, and hearing, which influences their ability to hear; and the course material is in the English language.

The procedure followed to achieve the prototypical instantiation of the model incorporated Canvas as a learning management system, which was utilized to create a Java Programming course. The experiment will be conducted using the Java course in the prototype. Mastery Paths was installed in Canvas as an adaptive learning tool to enable adaptive assessment. The Google Dialogflow platform was also used as the backend tool to create the conversational AI bot, which the learners will later use. The Infusion framework was utilized to improve the accessibility of the material, enabling learners with vision impairments to access the content.

In addition, a database was designed to accommodate the further needs of academic institutions, enabling them to analyze students' interactions with the LMS and make advanced analytics and predictions, such as grade prediction. An audio-to-text scenario was also implemented to provide accessibility to the material for learners with hearing impairments. ChatGPT was employed to respond to any inquiries or interactions made by learners for which no specific intent was designed to respond. In other words, it was used as a fallback response tool. Lastly, the Google Dialogflow messenger interface was adopted as the learner interface. The technology used in the prototype is detailed in Table 2.

The above prototypical implementation was administered and demonstrated in a workshop to a group of learners, instructors, and academic administrators. The experiment and the workshop indicated that it is not only about technology but pedagogy. Teaching must be conducted in an accessible and pedagogical manner. The material must be accessible and usable. Course design must be inclusive, as should teaching and instruction. Accessible and inclusive pedagogy and course design requires multiple means of engagement and representation. For ease of access and management, course content should be modularized. Technology should be employed to facilitate learners' accessibility. Academic institutions must have the following roles actively engaged: operator, instructor, designer, and learner. Finally, the technologies mentioned require optimization to reduce their footprint, costs, and administrative duties.

V. DESIGN PATTERNS

Designing innovative IT is a challenge to researchers and developers alike. Design knowledge must be codified and made available to support these individuals during the design process of such artifacts. Design patterns provide a working solution to address recurring problems in system development. They contain templates for describing design knowledge, often in tabular form. Design patterns have helped transfer the tacit knowledge of designers into explicit knowledge [28]. These patterns play a critical role in both academia and industry. For researchers, patterns provide a method for explicating design knowledge, while for practitioners, design patterns offer a solution to recurring problems. By applying them in a particular context, patterns represent elements of both the problem space and the solution space, providing an opportunity to bridge the gap between them. Owing to the abstraction of design patterns, they can be utilized for various application scenarios.

In system development, design patterns represent a solution to address recurring problems. Design patterns are established tools for making design knowledge accessible to other developers. The merits of patterns are that they cover various forms of knowledge (know-how, know-why, know-what) while providing a trusted approach to development practice [28]. Patterns are often referred to as ''templates' or ''best practices' for established solutions to frequently recurring problems in system development [29].

Design patterns are abstract and often represent reusable solutions for recurring problems. ''Best practices' are codified in design patterns and made usable for the future. Accordingly, a design pattern functions as a ''template' type, structured similarly for all patterns. These patterns do not present innovative solutions or contribute toward building solutions based on proven artifacts. A design pattern offers

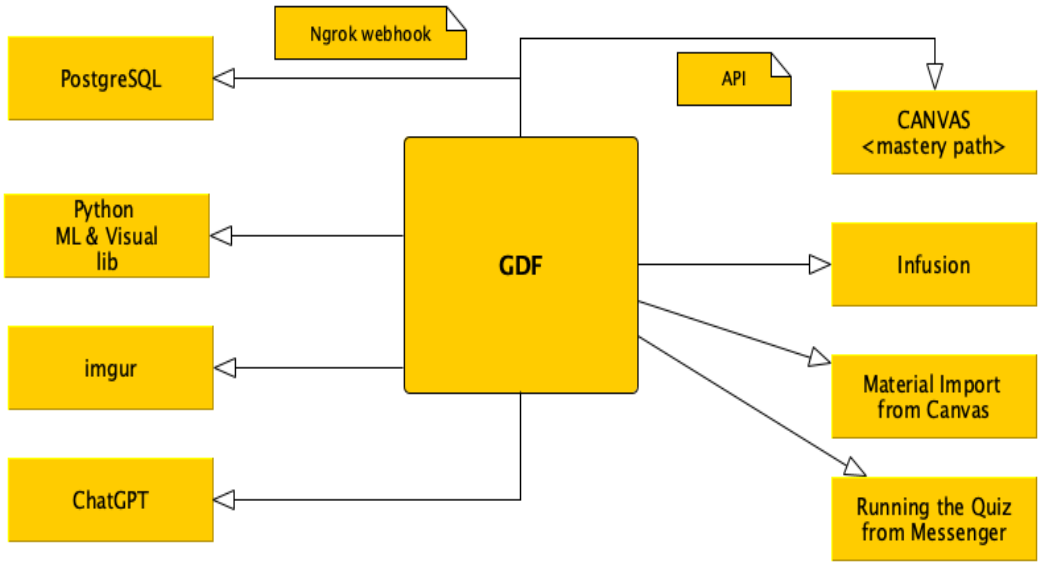


FIGURE 4. Evaluation environment.

TABLE 2. Technologies used in the proposed chatbot.

Technology	Purpose
Google Dialogflow	To create and administer the conversational bot.
Messenger	To act as the interface with which learners access course material.
PostgreSQL	Is the local database on which different learner master and transactional data are hosted and maintained.
Cloud DB	Is the same as previously administered but in the cloud rather than locally.
Canvas	The learning management system on which course material is created and instructors manage.
Canvas API	The API was utilized to cater to the integration needs of the other technologies in the prototype, e.g., the Infusion framework.
Mastery Paths	It has been utilized as an adaptive learning tool.
Python	Python code was written to facilitate integrating Google Dialogflow and other technologies, e.g., the PostgreSQL database engine.
ML Libraries	Machine learning libraries are utilized to make predictions, e.g., grade prediction.
Infusion framework	The infusion framework is applied to make course material accessible to those with vision severity.
JavaScript	This is the code that is employed to configure the Infusion framework.
Ngrok	It is used as the webhook server to facilitate integration.
OpenAI Whisper audion-to-text	The Whisper framework is applied to make course material accessible to those with hearing severity.
Imgur	It was utilized to store the plotly dashboards in the form of images.
Plotly library	It was employed to connect to the PostgreSQL database engine to create dashboards.
ChatGPT	When learners ask questions or make comments that Google Dialogflow cannot answer, the questions are forwarded to ChatGPT.

a path for addressing multiple problems. Therefore, design patterns address one of the problems identified by [30] in the reuse of design knowledge by codifying design knowledge in an abstract way [31].

TABLE 3. Design patterns focusing on the deployment of AI for efficient learning.

Use of AI conversational bots for efficient learning at academic institutions		
Goal: The overarching goal is to use AI conversational system for learning. Severely challenged learners benefit from the use of AI in education for inclusion. Overall, the learning environment is more efficient and capable of providing opportunities for all learner types. Both instructors and learners can engage in the learning mediated by the system.		
Requirements: In order to increase the overall quality of learning offered at academic institutions, the below are the requirements: 1) Design a learning environment that can offer inclusion for severely challenged learners. 2) Design an adaptive learning environment.		
Key Components:	**Integration**	**Roles**
— Learning management system	— All components need to be seamlessly integrated	— Instructor
— Conversational bot platform	— Top-up features e.g., prediction and dashboards are rendered as by-products	— Learner
— Database system	—	— Designer
— Visualization tool	—	— Operator
Solutions: The solution to the proposed problem must at least include a learning management system and an AI conversational bot platform. In addition, to provide added features such as analyzing logs and predicting academic performance, integration into machine learning prediction algorithms is required together with a visualization tool. This study proposes a tool to enable the integration via a secure and uniform integration platform. The minimum required functionality is: 1) The ability to upload course material to a learning management system and access it from an AI-enabled conversational bot. 2) Assessment can also be rendered via the conversational bot. 3) The bot enlivenment extends the learning management system's functionality and availability. 4) Such an environment needs access to several technologies and systems. Presently, this study orchestrates the integration via a cloud-based API gateway.		

TABLE 4. Design patterns focusing on the deployment of AI for efficient learning.

Use of AI conversational bots for efficient learning at academic institutions
Important Aspects: 1) Strive for a generic design: Since academic institutions are different, design the process to be generic to accommodate various needs. 2) Conversational AI Bots help academic institutions to be proactive: conversational bots require maintenance and analysis of logged activities and learners' logs. Such data need analysis and actions to help academic institutions become proactive over time. 3) The roles must be active: all parts should get involved adequately, as the adoption of conversational AI bots requires them to be active and provide inputs. Also, adjust based on the feedback and the change in technological aspects. 4) The design of the course is more important than the technology: the design of the course influences the use of technological aspects and features. Therefore, design needs to be put ahead of the technology. 5) Integration is an important aspect: for the learning environment to work efficiently, solid integration is required owing to the fact that it relies on several tools and technologies. 6) Organizational change management: while the technology utilized by the designed IT artifact enhances the learning environment at academic institutions, it calls for organizational change. For example, instructors play new roles in managing and interacting with the technology and its supporting components. Changes must be managed.

Designers of new technologies gain design knowledge through experience, education, and problem-solving skills. Although the amount of design knowledge a designer possesses is influenced by experience and exposure, the existing design knowledge of a developer can be termed tacit knowledge. This knowledge must first be made available to others via externalization, which can be done orally or in writing through codification [28]. Design patterns externalize tacit design knowledge by codifying it for others and recording it in written form. With the help of externalized design knowledge, design problems can now be solved. Although experienced developers have access to a wealth of experience, inexperienced developers usually have significantly fewer ''best practices.' Due to their abstract representation, design patterns do not provide a direct recipe for solving problems but support users in finding a suitable solution for the problem at hand by pointing out possible solution approaches. Thus, a design pattern can solve many problems in the field [29]. It is worth noting that for many problems, there is no single correct solution; instead, multiple solutions can satisfactorily solve the same problem. Vom Brocke et al. [30] indicated that patterns are a component of the design knowledge base. They help in finding suitable solutions to existing problems.

Table 3 and Table 4 show the study design pattern for using AI conversational bots for efficient learning. The design

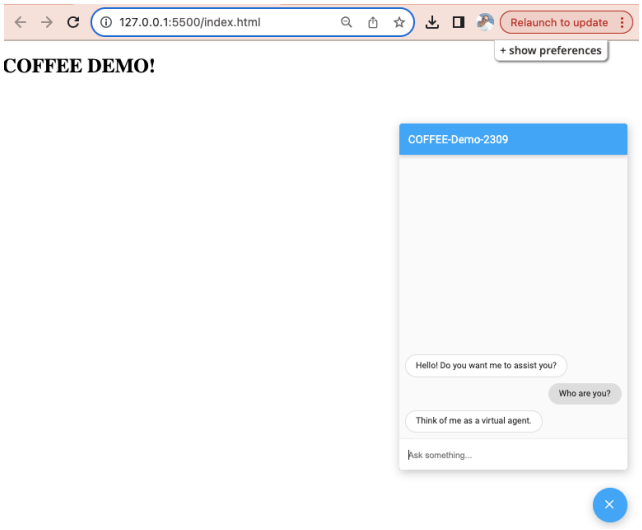


FIGURE 5. The main window of the chatbot after successful implementation.

pattern contains elements such as the goals to achieve, the requirements to address, the processes needed, the environment to support, the solution, and other essential aspects. The proposed design pattern can be easily instantiated by other designers when addressing efficient learning in academic institutions, regardless of the type of learning to support, the level of environment-specific parameters, or the nature of the learning material.

VI. RESULTS AND VALIDATION

This section presents the results and validates the successful implementation and integration of various technologies. First, creating the Graphical User Interface (GUI) for the chatbot's main windows was accomplished, signifying the successful realization of the initial design concept. The obtained chatbot GUI is illustrated in Figure 5.

The validation process continued to verify the seamless integration of the infusion framework. Figure 6 indicates that the infusion framework is effectively incorporated into the chatbot. In addition, various functionalities, including font size, background color, and line spacing, are thoroughly tested and validated. Figure 6 shows the chatbot's suitability for students with visual impairments, aligning with one of the primary design objectives.

The core functionality of the chatbot includes integrating with the Canvas LMS to create and manage assignments. Figure 7 demonstrates the successful integration with Canvas LMS, displaying a variety of pre-designed questions linked to the chatbot. The chatbot's decision-making process for selecting the next question is based on the accuracy of the student's previous answer. This personalized approach is designed to enhance the learning experience by providing students with a customized set of questions designed to their individual knowledge levels.

Integrating the chatbot with ChatGPT is a strategic decision to enhance its capability to respond to a broader range

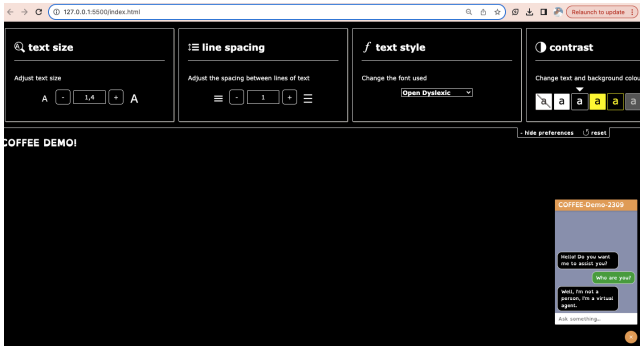


FIGURE 6. Implementing the infusion framework exhibits different options for students with special needs.

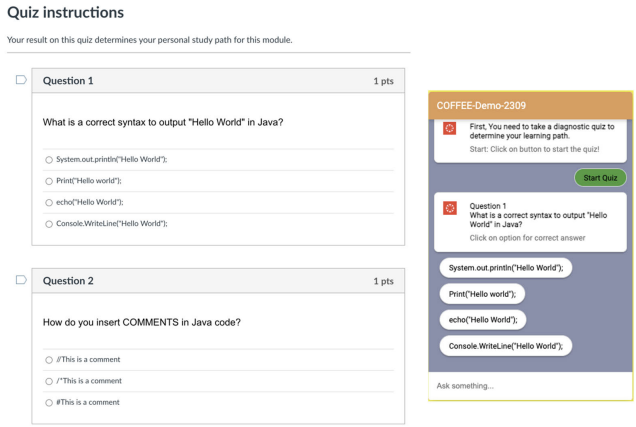


FIGURE 7. Implementing Canvas LMS integration for a particular course shows the possibility of efficient learning.



FIGURE 8. The chatbot is successfully integrated with ChatGPT to answer general questions outside the scope of the course.

of general inquiries, extending beyond the scope of course materials. Figure 8 displays the successful implementation of the ChatGPT integration. It indicates distinct background colors and fonts, serving as visual evidence of the effective operation of the infusion framework.

VII. DISCUSSION

Artificial intelligence can perform complex tasks that often require human intelligence. AI has made significant progress in education, offering innovative methods to enhance

teaching and learning. Conversational AI-enabled bots are employed to serve educational goals. Education must address learners' differences through adaptive learning settings, which is one of the study's guiding principles. Learners vary, and technology must accommodate their differences; therefore, a framework is utilized to make educational material accessible, aligning with the second study principle. In addition, educators are often overwhelmed with various tasks, and the availability of technology can assist them in providing a better learning environment.

The research is limited to instantiating the conceptual framework on the Google Dialogflow bot environment, Canvas, as a learning management system, and other supporting technologies. The prototypical implementation is also limited to a single course with a restricted material set. Although no profound changes are expected when implementing the design in other technological and environmental contexts, the impact must be studied accordingly.

VIII. CONCLUSION

This study introduces the design and implementation of a bot prototype capable of offering an adaptive learning environment and providing personalized access to educational material. Educational institutions have increasingly employed AI technologies to address education and learning challenges. The study demonstrates how AI can be utilized and integrated into learning management systems, specifically conversational bots. A fully integrated environment is planned for the future, where robotic process automation will automate the integration between different systems. In addition, there are plans to process more types of course material, such as audio and video. It is also recommended that AI be used for inclusion; therefore, all learning materials should be searchable through various search criteria and options. Support for different languages needs further investigation. The impact of integrating ChatGPT with the chatbot requires additional research and investigation. Specifically, to ensure the relevance of responses to prompts, further exploration of Retrieval-Augmented Generation (RAG) is planned on top of a Document Vector Database (DVDB) to help attain relevant and up-to-date responses.

DATA AVAILABILITY

No dataset was used for the research described in this article.

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